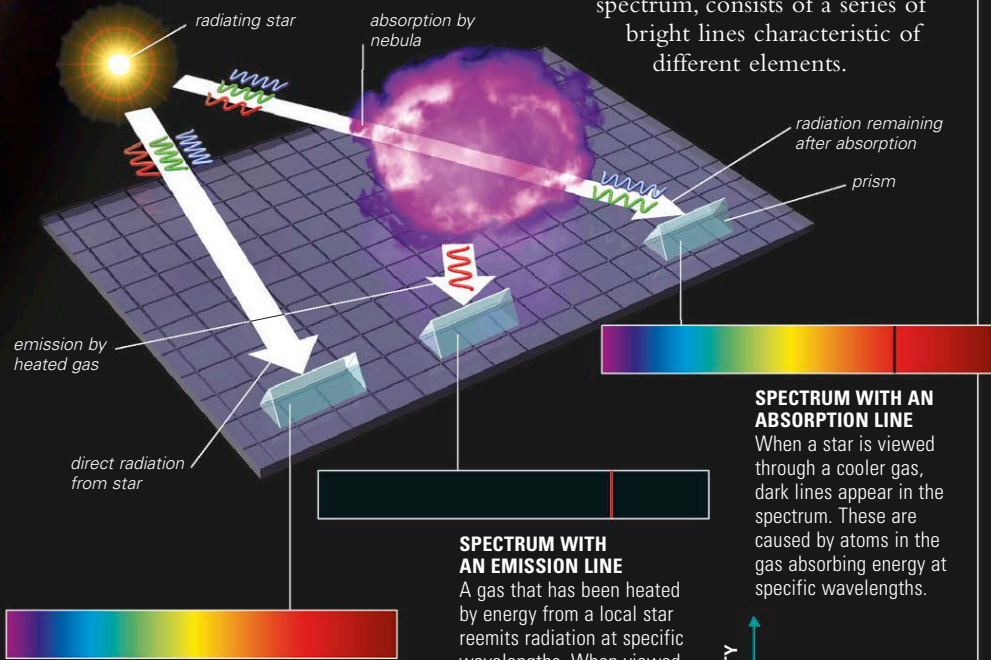


ANALYZING LIGHT

The radiation output of celestial objects is a mixture of wavelengths. When passed through a prism, the light is split into its component wavelengths, giving a record called a spectrum. A star's spectrum usually contains dark lines called absorption lines, caused by photons being absorbed at certain wavelengths by atoms in the star's atmosphere. They can be used to establish what chemical elements are present. The spectrum of a nebula can also reveal its composition. When heated by radiation from a nearby star, its atoms emit their own light. The resulting spectrum, called an emission spectrum, consists of a series of bright lines characteristic of different elements.



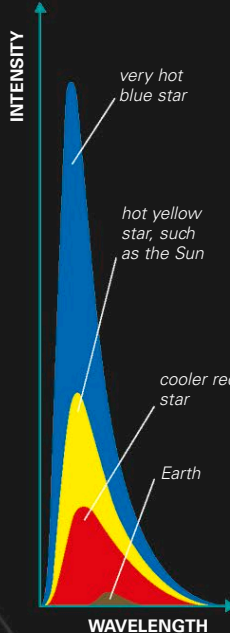
EMISSION NEBULA
This nebula glows as its gas is heated by nearby stars. The emitted light consists of photons of a few specific wavelengths. These photons were emitted by the gas's atoms as their electrons settled to lower energy states.

CONTINUOUS SPECTRUM
A hot, dense gas such as a star produces a continuous light spectrum from its surface, with all different light wavelengths (colors) represented.

SPECTRUM WITH AN EMISSION LINE
A gas that has been heated by energy from a local star reemits radiation at specific wavelengths. When viewed obliquely, this produces an emission-line spectrum.

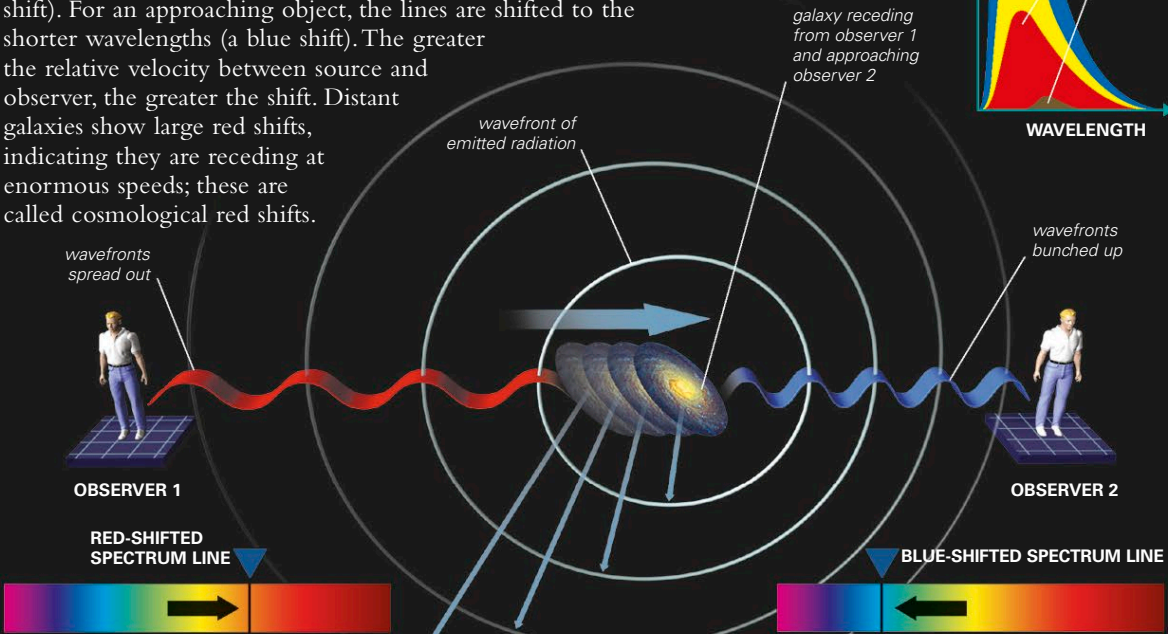
SPECTRUM WITH AN ABSORPTION LINE
When a star is viewed through a cooler gas, dark lines appear in the spectrum. These are caused by atoms in the gas absorbing energy at specific wavelengths.

RADIATION FROM HOT OBJECTS
Not only is the total radiation greater from hotter objects, but the wavelength of peak intensity is also shorter (toward the blue end of the light spectrum). Astronomers can calculate the temperature of stars by measuring the peak of the star's spectrum.



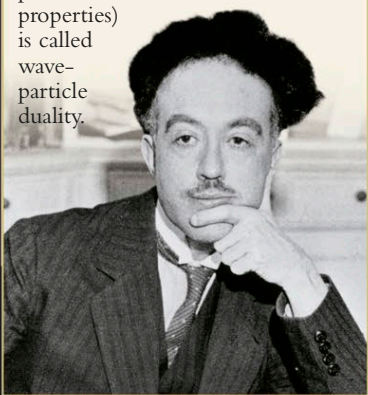
RED SHIFT AND BLUE SHIFT

The spectrum of radiation received by an observer shifts if the source of the radiation is moving relative to the observer—and celestial objects are always moving. Astronomers can detect the shifts by measuring the position of spectral lines, which occur at characteristic places. Observers watching an object moving away see its spectral lines shifted toward longer wavelengths (a red shift). For an approaching object, the lines are shifted to the shorter wavelengths (a blue shift). The greater the relative velocity between source and observer, the greater the shift. Distant galaxies show large red shifts, indicating they are receding at enormous speeds; these are called cosmological red shifts.



LOUIS DE BROGLIE

The French physicist Louis de Broglie (1892–1987) received the Nobel Prize in 1929. He found that particles of matter, such as electrons, have wavelike properties. The dual nature of matter and light (each has both particlelike and wavelike properties) is called wave-particle duality.



SHIFTING WAVELENGTHS
Shifts occur because of a phenomenon called the Doppler effect. The wavefronts of light from a receding object are stretched out, increasing their wavelength, while those of an approaching object are squashed up.